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Policy Evolution and Technological Breakthroughs in China's Deep-Sea Floating Offshore Wind Development: A Strategic Pathway to Commercialization

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Abstract

Against the backdrop of the global energy transition, China is accelerating the shift of its offshore wind development focus towards deep-sea areas. This paper systematically interprets the strategic leap in China's deep-sea offshore wind policy from the "dual-ten" rule to the "single-thirty" rule and its driving mechanisms. It further analyzes the core technological progress and challenges in floating wind technology. Research indicates that policy innovations have effectively unlocked over four times the resource potential of deep-sea areas. However, floating wind technology still faces bottlenecks in high costs, environmental adaptability, and industrial chain coordination. Future efforts require lightweight design, multi-energy complementary integration, and coordinated policies and standards to accelerate the large-scale, cost-competitive development of deep-sea offshore wind. This will provide core support for China's "Dual Carbon" goals and the global ocean energy transition.

Introduction

With the advancement of the "Dual Carbon" goals (carbon peak and carbon neutrality), China's offshore wind development focus is rapidly shifting towards deep-sea areas. Between late 2024 and early 2025, the Chinese government released the *Guidance Catalogue for Natural Resource Elements Supporting High-Quality Industrial Development-2024 Edition* (hereinafter referred to as the "Guidance Catalogue") and the *Notice on Further Strengthening Sea Area Management for Offshore Wind Projects* (hereinafter referred to as the "Notice"). These documents mark a historic transition in China's offshore wind policy framework from near-shore constraints to deep-sea exploration. This paper analyzes the strategic pathways, core challenges, and future directions of China's deep-sea offshore wind development by examining policy evolution and floating technology advancements.

Evolution of Deep-Sea Offshore Wind Policy: Strategic Leap from "Dual-Ten" to "Single-Thirty"

Foundation and Limitations of the "Dual-Ten" Era (2011-2024)

The 2011 *Detailed Implementation Rules for the Interim Measures on Offshore Wind Power Development and Construction Management* established the "dual-ten" standard (offshore distance ≥ 10 km and water depth ≥ 10 meters), facilitating the large-scale development of near-shore projects like **Jiangsu Rudong**. However, intensified competition for near-shore resources and ecological conflicts increasingly constrained the available development space.

The CGN Rudong Project is located in the Yellow Sea offshore of Rudong County, Nantong City, Jiangsu Province, China. Construction officially commenced on May 8, 2015. The project comprises 38 wind turbine units and includes one offshore and one onshore substation, with a total installed capacity of 152 MW. Upon commissioning, the project is projected to generate approximately 400 million kWh of electricity annually. Compared to a coal-fired power plant, the Rudong project is estimated to save 131,000 tons of standard coal, reduce CO₂ emissions by 267,000 tons, decrease ash residue by 49,000 tons, and lower soot emissions by 13,000 tons each year. At the time (2016), the Rudong project was China's first fully operational offshore wind project with the farthest offshore distance and the largest installed capacity. Located approximately 25 kilometers from the coast, with seabed elevations ranging between -8 meters and -14.6 meters, it was also China's first offshore wind project compliant with the "Double Ten" standard (water depth > 10 m, offshore distance > 10 km).



Figure 1—CGN Jiangsu Rudong Offshore Wind Farm

Innovative Breakthrough of the "Single-Thirty" Policy (2024-)

The densely issued Guidance Catalogue for Natural Resource Elements Supporting High-Quality Industrial Development (2024) and the *Notice* in 2024-2025 signify a policy shift towards deep-sea areas:

- Spatial Layout Innovation:** Establishes the "single-thirty" standard (offshore distance ≥ 30 km or water depth > 30 meters), unlocking the deep-sea resource potential (theoretical capacity: 600 GW, 3-4 times that of near-shore areas).

- **Intensive Sea Use Mechanism:** Controls sea area usage per unit capacity (≤ 8 MW: ≤ 0.13 km²/MW; ≥ 12 MW: ≤ 0.15 km²/MW), improving sea area utilization efficiency by over 20% compared to the 2016 standard (0.16 km²/MW).
- **Centralized Approval Authority:** Projects requiring sea areas ≥ 700 hectares, spanning provincial boundaries, or located outside provincial jurisdictional waters require State Council approval, strengthening central ecological oversight.

Table 1—Core Innovative Mechanisms of Deep-Sea Offshore Wind Policies

Policy Dimension	Core Requirement	Breakthrough Significance
Spatial Layout	Established "single-thirty" standard, accommodating dual constraints of water depth and offshore distance	Guides projects towards deeper waters and farther shores, unlocking over 4 times the resource potential of deep-sea areas
Intensive Sea Use	Tiered sea area control per unit capacity (≤ 8 MW: ≤ 0.13 km ² /MW; ≥ 12 MW: ≤ 0.15 km ² /MW)	Improves sea area utilization efficiency by >20% compared to 2016 standard (16 ha/MW)
Sea Area Definition	Boundary defined by outermost edge of turbine blade sweep, replacing the previous "50m outward extension" model	Reflects actual sea occupation more accurately, reducing conflicts
Approval Authority	Projects ≥ 700 ha, cross-provincial, or outside provincial waters require State Council approval	Strengthens central coordination and ecological oversight, preventing disorderly local development
Market Mechanism	Implements "pre-cleared sea area" (Govt pre-clears area), explores sea area use right trading	Reduces upfront costs for enterprises, accelerates project initiation

Advances in Floating Wind Technology: From Demonstration to Commercialization Challenges

Demonstration Project Breakthroughs and Cost Challenges

As of 2024, demonstration projects such as "Three Gorges Leading" (5.5 MW) and "CNOOC Guanlan" (7.25 MW) have been commissioned, validating the feasibility of deep-sea power supply. However, high costs remain the core bottleneck:

- Unit cost $\approx 33,000$ yuan/kW (1.8 times that of fixed-bottom near-shore);
- Steel consumption for floaters ≈ 530 t/MW, significantly higher than the cost-competitive target (350 t/MW).

Table 2—Commissioned Floating Wind Turbine Demonstration Projects in China

Demonstration Project	Foundation Type	Quantity	Unit Capacity (MW)
Three Gorges Leading	Semi-submersible	1	5.5
CSSC Fuyao	Semi-submersible	1	6.2
CNOOC Guanlan	Semi-submersible	1	7.25
CHN Energy Sharing	Semi-submersible	1	4.0
MingYang Tiancheng	Semi-submersible	1	8.3

Three Gorges Leading

The "Three Gorges Leading" represents the world's first complete set of typhoon-resistant floating offshore wind power generation equipment, featuring a single-unit capacity of 5.5 MW. Its major components include the floating wind turbine unit, tower, three-column semi-submersible foundation platform, mooring system,

and dynamic cable. The platform is designed to withstand extreme wind, wave, and current conditions corresponding to a 50-year return period, capable of resisting typhoons up to Category 17. It integrates comprehensive monitoring systems, including wind turbine monitoring, motion attitude monitoring, stress monitoring, corrosion monitoring, liquid level monitoring, mooring system and dynamic cable monitoring, and marine environmental monitoring, all accessible remotely. This project pioneered the application of integrated analysis technology for the floating wind turbine-foundation platform-mooring system and dynamic cable in China. It overcame key technical challenges in designing mooring systems and dynamic cables for floating wind turbines under shallow water effects.



Figure 2—Three Gorges Leading FOWT

The project completed the R&D and design work for the floating wind turbine platform and mooring system, developing a novel typhoon-resistant floating foundation tailored to the marine environment of the South China Sea. Collaborating with industry chain partners, it successively tackled critical technologies in floating offshore wind power, including typhoon-resistant floating wind turbines, ballast control, dynamic cable development, and integrated towage and positioning installation of the "turbine + platform". The core solutions primarily utilize proprietary Chinese technology. By early 2022, the "Three Gorges Leading" achieved full-load operation at 5.5 MW. To date, the floating unit maintains an average availability rate of 98% and a capacity factor of 0.33, generating an average annual grid-connected electricity of 15.818 GWh. Assuming a standard coal consumption rate for thermal power of 301.5 g/kWh, the "Three Gorges Leading" saves approximately 4,800 tons of standard coal annually after commissioning. This contributes significant economic benefits through coal savings and increased electricity generation, playing an important role in achieving carbon peak and carbon neutrality goals and promoting the comprehensive green transformation of economic and social development.

CSSC Fuyao

The "CSSC Fuyao" is China's first domestically developed deep-sea floating offshore wind power equipment, spearheaded by CSSC Haizhuang Windpower Co., Ltd. This unit is equipped with a 6.2 MW typhoon-resistant wind turbine generator. The floating platform has an overall length of 72 meters, a width of 80 meters, a depth of 33 meters, and a total weight exceeding 4,000 tons [3–6]. It is deployed in the Luodousha sea area near Zhanjiang's Xuwen County, with an average water depth of 65 meters. Utilizing a semi-submersible design and a catenary mooring system, it can withstand sea conditions with a 50-year return period. The "Fuyao" achieves 100% domestic sourcing for primary components and over 95%

domestic sourcing for component-level parts. It fills the gap in China's deep-sea floating wind power technology and contributes to achieving the "Dual Carbon" goals.



Figure 3—CSSC Fuyao FOWT

CNOOC Guanlan

The "CNOOC Guanlan" is China's first deep-sea floating wind power platform, with an installed capacity of 7.25 MW, consisting of a floating foundation and wind turbine. This platform is the world's first semi-submersible deep-sea wind power platform deployed at the greatest depth and distance, and also the first globally to supply power to an offshore oil and gas field under the harshest marine environmental conditions. It is installed in an offshore oilfield area 136 kilometers from Wenchang, Hainan Province.



Figure 4—CNOOC Guanlan FOWT

On January 2, 2023, the main structure of the "CNOOC Guanlan" was completed in Qingdao. On February 23, 2023, it was successfully towed from Qingdao to Zhuhai, covering a total distance of nearly 1,420 nautical miles over approximately 15 days. Subsequently, the platform was transported from Zhuhai to the designated offshore oilfield area for installation. On March 26, 2023, the "CNOOC Guanlan" departed from the Fluor terminal in Zhuhai, Guangdong, heading towards the waters off Wenchang, Hainan. By

May 6, 2023, the "CNOOC Guanlan" successfully established the transmission "artery" connecting the Wenchang oilfield cluster, meeting the conditions for grid connection, signifying imminent commissioning. On May 20, 2023, the "CNOOC Guanlan" was successfully integrated into the Wenchang oilfield cluster grid, officially supplying green electricity to the offshore oil and gas field.

With an installed capacity of 7.25 MW, the "CNOOC Guanlan" is installed 136 kilometers offshore from Wenchang, Hainan. It achieves internationally advanced levels in multiple indicators, including investment per megawatt, steel consumption per megawatt, and single-unit floating turbine capacity. After commissioning, the electricity generated by the turbine is fed into the offshore oilfield cluster grid via a dynamic submarine cable for oil and gas production. The platform is installed in a sea area characterized by high winds and waves, making it the world's first "dual-hundred" offshore wind project (water depth >100 meters, offshore distance >100 kilometers).

CHN Energy Sharing

The "CHN Energy Sharing" platform was constructed within the National Marine Ranch Demonstration Zone on Nanri Island, Putian, Fujian Province. It represents the world's first integrated design project combining deep-sea floating wind power with aquaculture. The platform primarily consists of a "floating wind turbine + aquaculture net cage," including a floating platform, a 4 MW wind turbine, and a mooring and anchoring system. The floating platform features a three-column semi-submersible structure shaped like a regular triangular prism, with a column spacing of 70 meters, column height of 28 meters, and a design draft of 14 meters. The aquaculture water volume is approximately 10,000 cubic meters. Operating at full load, the platform's 4 MW wind turbine can generate 4,000 kWh per hour, translating to 96,000 kWh per day – equivalent to the daily electricity consumption of 42,500 people. On April 2, 2024, the platform released its first batch of 10,000 large yellow croaker fry, each weighing 700-800 grams, for a cultivation cycle of 3 to 4 months.



Figure 5—CHN Energy Sharing

Leveraging a new energy digital platform, the "CHN Energy Sharing" project pioneered the integrated multi-functional monitoring of "floating wind power + deep-sea aquaculture," enabling remote monitoring and unmanned operation. The intelligent deep-sea aquaculture equipment allows for remote visual observation of underwater fish schools and assessment of fry growth status, size, and weight. The electricity generated by the offshore wind turbine ensures the safe and stable operation of the aquaculture equipment, addressing the common issues of "insufficient and unstable power supply" in traditional marine ranches, thereby realizing "using wind power to support aquaculture and vice versa." The project marks the first

application of materials such as ultra-high molecular weight polyethylene fiber, fluorinated film, and hull-end connection tensioners in the floating wind power sector, promoting equipment upgrades in both marine aquaculture and offshore wind power industries.

MingYang Tiancheng

The "MingYang Tiancheng" is the world's largest single-unit floating wind power platform by capacity. It is projected to generate approximately 54 million kWh annually, sufficient to meet the annual electricity needs of 30,000 three-person households. Installation of the "MingYang Tiancheng" was formally completed in Nansha, Guangzhou on July 3, 2024. It was officially commissioned in Yangjiang, Guangdong on December 11, 2024.

The "MingYang Tiancheng" features two towers arranged in a "V" shape, supporting two 8.3 MW offshore wind turbines, resulting in a total capacity of 16.6 MW. It is designed for deployment in waters deeper than 35 meters globally. The highest point of the rotor reaches 219 meters, with a maximum rotor diameter of approximately 369 meters. The entire wind power platform has a total displacement of about 15,000 tons and an overall towage design draft of 5.5 meters.

The platform comprises a "Y"-shaped floating foundation on the sea surface and a "V"-shaped tower structure mounted at the central connection point of the "Y". The floating foundation measures 117 meters long, 105 meters wide, and 39.5 meters high, with a total weight exceeding 8,000 tons, forming the critical structure for the platform's stable operation in deep seas. The "MingYang Tiancheng" uniquely integrates two turbines on a single platform, a global first. Each turbine has a capacity of 8.3 MW, combining for a total capacity of 16.6 MW, making it the world's largest single-unit floating wind power platform.



Figure 6—MingYang Tiancheng FOWT

The two turbines stand side-by-side with their rotors turning in opposite directions. This design increases the wind speed in the central area between the rotors, thereby enhancing the conversion of kinetic energy from the air into electricity. Compared to a single large rotor with the same swept area, this configuration boosts power generation by 4.29%.

The "MingYang Tiancheng" incorporates several world-first innovations. Its floating foundation is the first globally to be constructed using ultra-high-performance concrete (UHPC) with a compressive strength exceeding 115 MPa, approximately four times stronger than conventional concrete. The pontoon design, utilizing a "glass fiber shell + XPS core material + protective coating material" scheme, is also a world-first, breaking away from the limitations of traditional pure steel or steel-concrete structures. Furthermore, the innovative design elements – including the winged twin towers, dual turbines, counter-rotating blades,

and the application of a cable-stayed system – are all global pioneers. This design is expected to increase power generation by 4.29% and enables the tower structure to assist in wind alignment.

Key Technological Bottlenecks and Innovation Directions

Enhancing Environmental Adaptability.

- **Floater Optimization:** Semi-submersible dominates (74%); Tension Leg Platforms (TLP), with superior motion stability and lower steel consumption (31% reduction vs. semi-sub), are a key focus for lightweighting.
- **Mooring Innovation:** Suction anchors offer 40% lower cost than pile anchors in 100m water depth; shared mooring systems can reduce anchor count by 42%.

Industrial Chain Localization Breakthroughs.

- **Core Equipment:** Breakthroughs in domestic production of synthetic fiber mooring lines and dynamic cables; development of TLP-specific tensioners.
- **Engineering Software:** Development of independent coupled design platforms, validated alongside third-party tools (e.g., DNV Sima).

Floating wind project utilizing TLP foundations. TLPs (Tension Leg Platforms) rely on tension tendons for stability, making them sensitive to water depth conditions. Shallow water areas may increase platform motions and structural stresses. The stability of TLPs is highly dependent on their tension tendons. Due to the complexity of their transportation and installation processes, where the tendons are not yet connected, temporary buoyancy modules are required to ensure safe operations during these phases.

In June 2025, the Provence Grand Large (PGL) project, developed jointly by EDF and Enbridge, achieved full-capacity grid connection. As the world's first deep-water floating wind project utilizing TLP foundations, this marks a significant breakthrough in the application of this innovative technology within the wind energy sector.

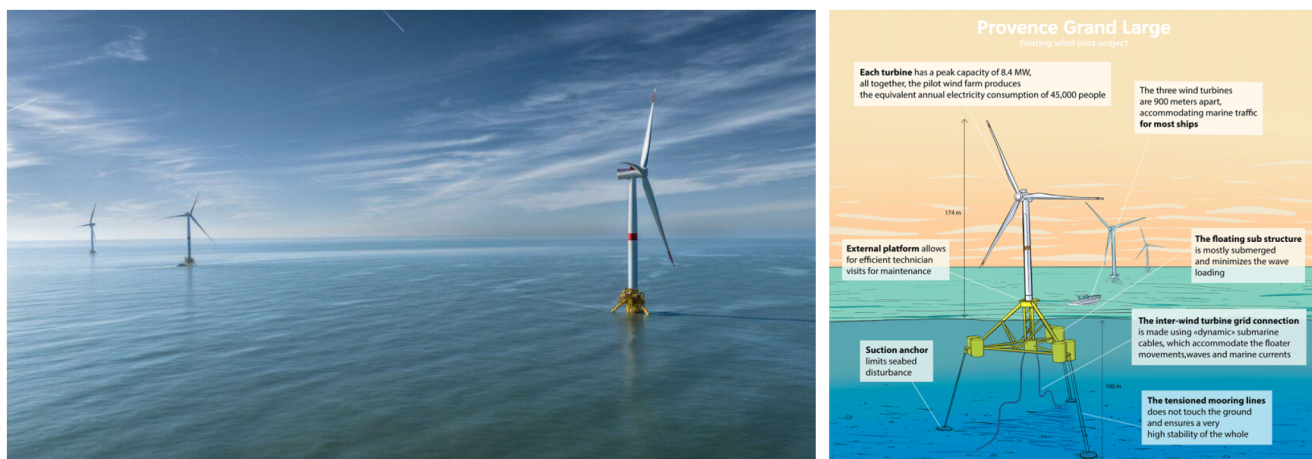


Figure 7—Provence Grand Large- floating offshore wind farm

On February 19, 2025, the construction of the floating structure for the CNOOC Lufeng Oilfield Group Clean Energy Power Supply Renovation Demonstration Project officially commenced at its fabrication yard in Qingdao. This signifies the formal entry into the engineering phase of the world's largest TLP floating wind demonstration project by single-turbine capacity.

Located in the sea area off eastern Guangdong, 136 km offshore at a water depth of 136 meters, this project represents China's first deep-water wind power demonstration project employing TLP technology. It

will deploy the world's largest 16 MW floating wind turbine units. For the first time, it innovatively adopts a vertical tension leg mooring system, integrating dynamic and static submarine cables along with energy storage devices to supply power to the Lufeng oilfield group.

This project integrates research and engineering. It represents CNOOC's further exploration and practical application of deep-water floating wind technology following the "CNOOC Guanlan" project. Developed entirely through CNOOC's independent innovation, the project has overcome core technical challenges including TLP floating platform design and construction, mooring system design and development, fully autonomous offshore installation, and the large-scale integration of energy storage within oilfield microgrids. Multiple project metrics fill domestic technological gaps, achieving a leap from technological catch-up to partial leadership in the deep-water wind power domain.

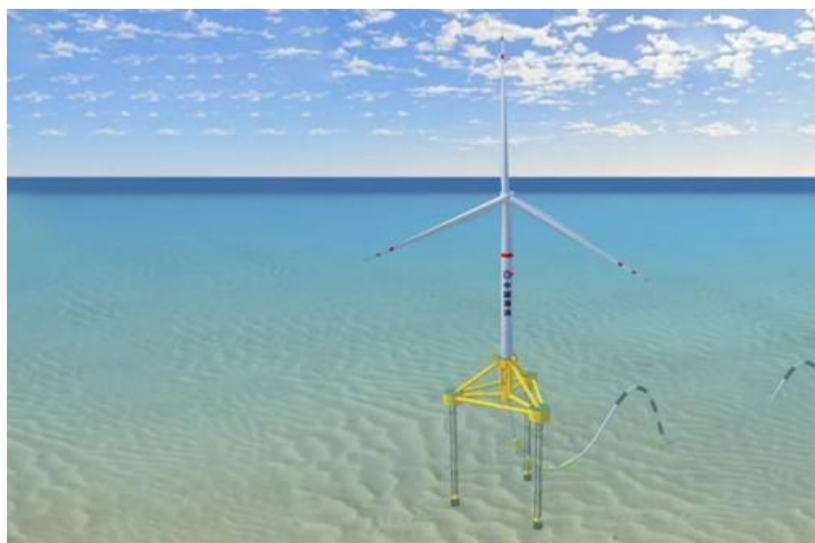


Figure 8—COOEC FWOT Platform of TLP Type

Compared to other floating wind turbine technologies, TLP technology offers distinct advantages in sea space utilization. Its motion response characteristics are closer to those of fixed-bottom turbines, significantly improving the operational environment for floating turbines. This leads to enhanced turbine reliability and increased power generation. Furthermore, its floating structure is simpler, offering significant advantages in steel consumption per kilowatt (kW), facilitating large-scale serial production.

Multi-EnergyComplementary Integration Models

- **"Wind + Oil/Gas + Hydrogen" Synergy:** Utilizing offshore oil/gas pipeline networks for surplus electricity hydrogen production (e.g., European AquaVenus project).

"Wind Power Coupled with Seawater Electrolysis for Hydrogen Production" represents a critical pathway for addressing power consumption challenges associated with far-reaching offshore wind farms at the current stage. European energy giants have targeted offshore hydrogen production, having already launched the pilot demonstration project "AquaVentus" in Germany as a forerunner initiative. China, possessing well-developed subsea pipeline networks in the South China Sea, has commenced preliminary research on localized consumption of floating wind power with surplus electricity directed towards hydrogen production. This groundwork lays the foundation for future large-scale, integrated exploitation of diverse energy forms – including offshore oil and gas, wind power, and hydrogen energy – in deep-sea areas.

- **Off-Grid Power Supply Model:** "CNOOC Guanlan" directly powers offshore oil/gas platforms, avoiding grid connection costs.

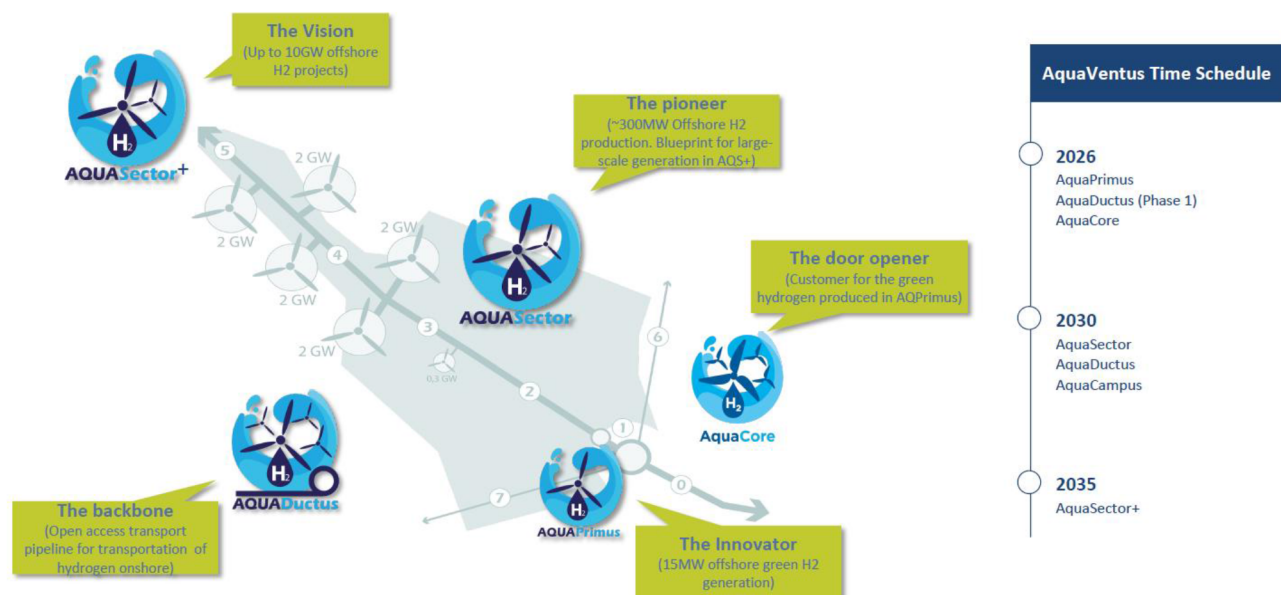


Figure 9—AquaVentus Project Family—Steps towards large-scale offshore H2 generation in 2030+

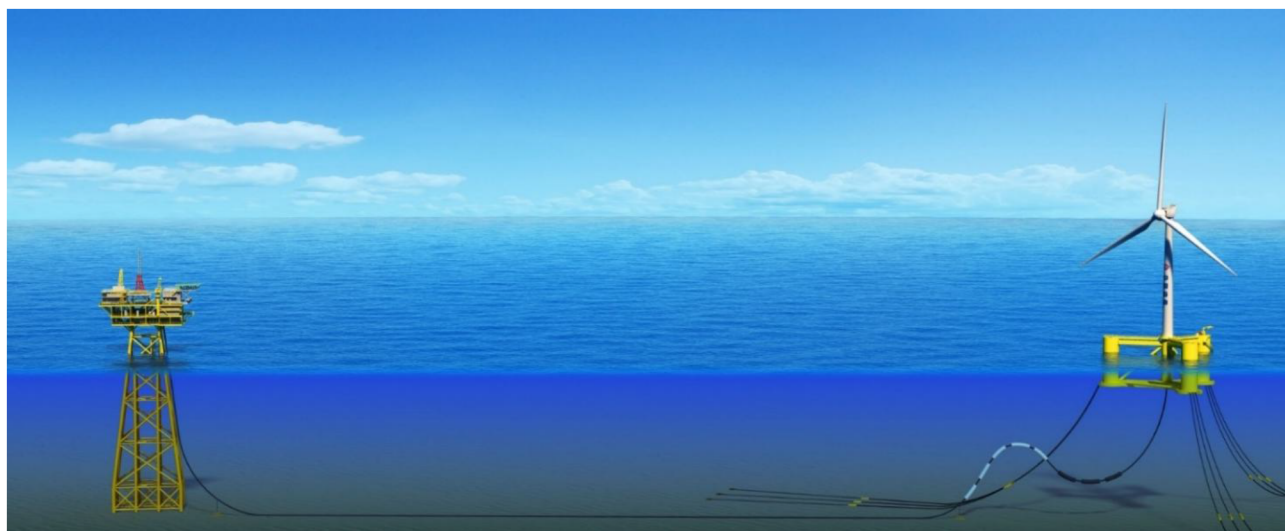


Figure 10—CNOOC Guanlan: FOWT+Oil/Gas Production

Challenges and Pathways for Policy-Technology Synergy

Current Development Bottlenecks

- **Economic Constraints:** Floating wind costs are ~3 times higher than near-shore; O&M costs for far-sea projects account for up to 20%.
- **Technical Risks:** Frequent typhoons in the South China Sea (wave height >12m) pose significant high-frequency fatigue risks to mooring systems.
- **Approval Barriers:** Lack of specific standards; complex cross-departmental approval processes.

Pathways for Policy-Industry Synergy Optimization

- **Optimize Top-Level Design:** Formulate *Design Guidelines for Floating Offshore Wind*; establish a cross-departmental "one-stop" approval platform.

- **Incentivize Technological Innovation:** Expand "wind power + fishery" integration demonstrations; set a 2030 target of 50% cost reduction for floating wind.
- **Integrate Industrial Chain:** Promote standardization of floater-turbine interfaces; strengthen R&D in weak areas like dynamic cables.

Table 3—Floating Wind Cost Reduction Pathways and Potential (2024 vs 2030 Target)

Technology Direction	Current Level	2030 Target	Cost Reduction Contribution
Floater Steel Use	530 t/MW	350 t/MW	25% reduction in foundation cost
Mooring System	Conventional Pile	Shared Suction	40% reduction in mooring cost
O&M System	Manned Vessel	Drone + Digital Twin	30% reduction in O&M cost



Figure 11—CHN Energy Sharing-"Wind+Fishery"

Outlook for Deep-Sea Offshore Wind: Trillion-Dollar Market and Global Leadership

Policies set a clear target of 200 GW installed offshore wind capacity by 2030, driving total industrial chain output value exceeding 20 trillion yuan. Floating wind, as the core enabler for deep-sea development, will dominate future technological iterations:

- **Turbine Upscaling:** Turbines >12 MW to match intensive sea-use requirements.
- **Intelligent O&M:** Digital twin platforms enabling predictive diagnostics.
- **Global Market:** China holds 64% of global blade and 80% of gearbox manufacturing capacity; deep-sea experience will be shared globally.

Conclusion

China's deep-sea offshore wind policy shift, from "dual-ten" to "single-thirty," marks a strategic reorientation towards deep-sea development. While floating wind technology has achieved breakthroughs in demonstration projects, cost reduction requires innovations such as lightweight floaters (TLP platforms), shared mooring, and multi-energy complementarity. On the policy front, expediting specialized standard formulation and approval reform is crucial. Technologically, breakthroughs in "bottleneck" areas like high-voltage dynamic cables and synthetic mooring lines are essential. Only through deep policy-industry synergy can large-scale, cost-competitive development of deep-sea offshore wind be achieved before 2030, contributing "China's Practice Recommendation" to the global ocean energy transition.

China's strategic pivot in offshore wind policy—from the restrictive "dual-ten" (offshore distance ≥ 10 km, water depth ≥ 10 m) to the enabling "single-thirty" standard (offshore distance ≥ 30 km or water depth > 30 m)—marks a decisive shift toward unlocking deep-sea wind resources, quadrupling exploitable capacity to 600 GW. While floating wind demonstration projects (e.g., Three Gorges Leading, CNOOC Guanlan, MingYang Tiancheng) validate technical feasibility, cost competitiveness remains the primary bottleneck. Current floating wind unit costs ($\sim 33,000$ CNY/kW) are $1.8\times$ higher than fixed-bottom projects, driven by excessive steel consumption (530 t/MW vs. 350 t/MW target) and complex O&M in harsh environments.

To accelerate commercialization, dual-track advancements are imperative:

a. Technological Innovation:

- Lightweight floaters (e.g., TLP platforms reducing steel use by 31% vs. semi-submersibles) and shared mooring systems (42% fewer anchors).
- High-voltage dynamic cables, synthetic fiber mooring lines, and multi-energy integration (e.g., CNOOC Guanlan's oilfield microgrid; AquaVentus-style "wind-to-hydrogen").
- Turbine upscaling (> 12 MW) and AI-driven O&M (digital twins) to cut operational costs by 30%.

b. Policy-Industry Synergy:

- Streamline approval via cross-departmental "one-stop" platforms and specialized standards (e.g., Floating Offshore Wind Design Guidelines).
- Incentivize hybrids (e.g., CHN Energy Sharing) and set cost-reduction targets (50% by 2030).

By 2030, China aims for 200 GW offshore wind capacity, catalyzing a > 20 trillion CNY (2.74 trillion USD) industrial chain. Achieving this requires resolving deep-sea challenges—typhoon resilience (wave heights > 12 m), supply chain localization, and grid integration—through policy-technology coordination. Success will position China as a global leader in floating wind, exporting expertise and equipment (64% of global blade capacity) while advancing the "Dual Carbon" goals.

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